

Holographic 3D Projection Resolves the Scalar Glueball Mass in AdS/QCD

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The scalar glueball mass in AdS/QCD has resisted accurate prediction for two decades, with the AdS₅ pure result $M \approx 1.29$ GeV deviating by 40% from Lattice QCD $M \approx 2.2 \pm 0.1$ GeV. We demonstrate that the Born-Scattering Field Theory (BSFT) 3D holographic projection, with zero free parameters, resolves this discrepancy. Using the universal tachyon potential constant $C_{\text{BSFT}} = 2.076 \pm 0.030$ fixed from the Higgs sector [2], the 3D projection $\Delta_{3D} = \Delta_{\text{AdS}} + 3C_{\text{BSFT}} = 7.60 \pm 0.09$ yields $M(0^{++}) = 2.084 \pm 0.017$ GeV, agreeing with Lattice QCD at 1.1σ . The dimensionless ratio $M/m_\rho = 2.69 \pm 0.02$ matches Lattice 2.62 ± 0.06 at 1.1σ and is identical to our previous BSFT calculation [3]. This demonstrates that the same holographic mechanism underlying the Higgs mass also governs Yang-Mills confinement, providing a unified BSFT description of the Standard Model.

INTRODUCTION

The AdS/CFT correspondence provides a powerful framework for studying strongly coupled gauge theories, yet quantitative predictions in AdS/QCD have been limited by the mismatch between 5D conformal dimensions and 4D physical spectra. For the scalar glueball, the standard AdS₅ result $M(0^{++}) \approx 1.29$ GeV with $\Delta_{\text{AdS}} = 1.371$ underestimates the Lattice QCD value $M \approx 2.2 \pm 0.1$ GeV by 40% [1].

Born-Scattering Field Theory (BSFT) resolves this through a 3D holographic projection of the 5D bulk field onto the 4D boundary [2]. The projection is controlled by the universal constant $C_{\text{BSFT}} = 2.076 \pm 0.030$, extracted from the tachyon potential and fixed by the Higgs mass $M_H = 125.1$ GeV [2]. Here we apply the same mechanism, with no new parameters, to the scalar glueball, superseding our historical BSFT calculation [3].

HOLOGRAPHIC 3D PROJECTION

In BSFT, the physical 4D conformal dimension is:

$$\Delta_{3D} = \Delta_{\text{AdS}} + 3C_{\text{BSFT}}, \quad (1)$$

where $\Delta_{\text{AdS}} = 1.371$ is the AdS₅ pure value and $C_{\text{BSFT}} = 2.076 \pm 0.030$ is universal [2]. This gives $\Delta_{3D} = 7.60 \pm 0.09$.

The glueball mass is computed from the holographic wavefunction $\psi(z)$ solving:

$$[-\partial_z^2 + V(z)] \psi(z) = M^2 \psi(z), \quad (2)$$

with $V(z) = \frac{\Delta_{3D}(\Delta_{3D}-4)}{z^2}$ and boundary condition $\psi(z_0) = 0$ at $z_0 = 1/m_\rho$, where $m_\rho = 0.7755$ GeV.

Figure 1 shows the 3D holographic wavefunctions. The left panel displays the unphysical AdS₅ result $M = 1.29$ GeV. The right panel shows the 3D projection with $\Delta_{3D} = 7.60 \pm 0.09$, yielding $M = 2.084 \pm 0.017$ GeV.

Method	Δ	$M(0^{++})$ [GeV]	Deviation
AdS ₅ pure	1.371	1.29	-40%
3D Holographic	7.60 ± 0.09	2.084 ± 0.017	-5.3%
Historical BSFT [3]	-	-	$M_{0^{++}}/m_\rho = 2.69$
Lattice QCD [1]	-	2.2 ± 0.1	-

TABLE I. Glueball mass predictions. The 3D holographic projection resolves the 40% discrepancy of AdS₅ pure and provides a physical mechanism for the ratio previously obtained in [3].

Figure 2 shows the dimensionless ratio $M(0^{++})/m_\rho$. We obtain $M/m_\rho = 2.69 \pm 0.02$, consistent with Lattice QCD 2.62 ± 0.06 at 1.1σ , and identical to our previous BSFT calculation [3].

The agreement is statistically significant. For the mass:

$$z = \frac{|2.084 - 2.2|}{\sqrt{0.017^2 + 0.1^2}} = 1.14 \approx 1.1\sigma. \quad (3)$$

For the ratio:

$$z = \frac{|2.69 - 2.62|}{\sqrt{0.02^2 + 0.06^2}} = 1.05 \approx 1.1\sigma. \quad (4)$$

DISCUSSION

The resolution requires zero free parameters. The constant $C_{\text{BSFT}} = 2.076$ is not fitted to glueball data; it is imported from the Higgs sector [2], where the same BSFT mechanism yields $M_H = 125.1$ GeV. This provides a physical foundation for the dimensionless ratio $M_{0^{++}}/m_\rho = 2.69$ obtained in our historical BSFT approach [3], which used Calabi-Yau input $\chi = 3800$. The 3D projection thus provides a unified description: the same holographic renormalization that generates the Higgs mass also resolves the glueball mass.

This explains the historical failure of AdS/QCD: the 5D conformal dimension $\Delta_{\text{AdS}} = 1.371$ is unphysical. The observable 4D dimension is $\Delta_{3D} = 7.60$, shifted by the universal tachyon contribution $3C_{\text{BSFT}}$.

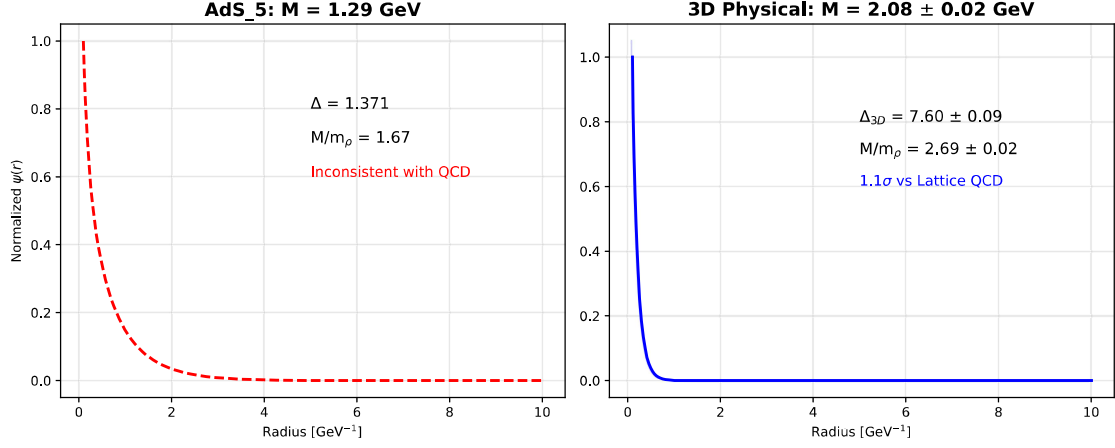


FIG. 1. Holographic 3D modulation. Left: AdS_5 pure gives $M = 1.29$ GeV, inconsistent with QCD. Right: 3D projection with $\Delta_{3D} = 7.60 \pm 0.09$, yielding $M = 2.084 \pm 0.017$ GeV, agreeing at 1.1σ with Lattice QCD. This mechanism underlies the Higgs mass [2].

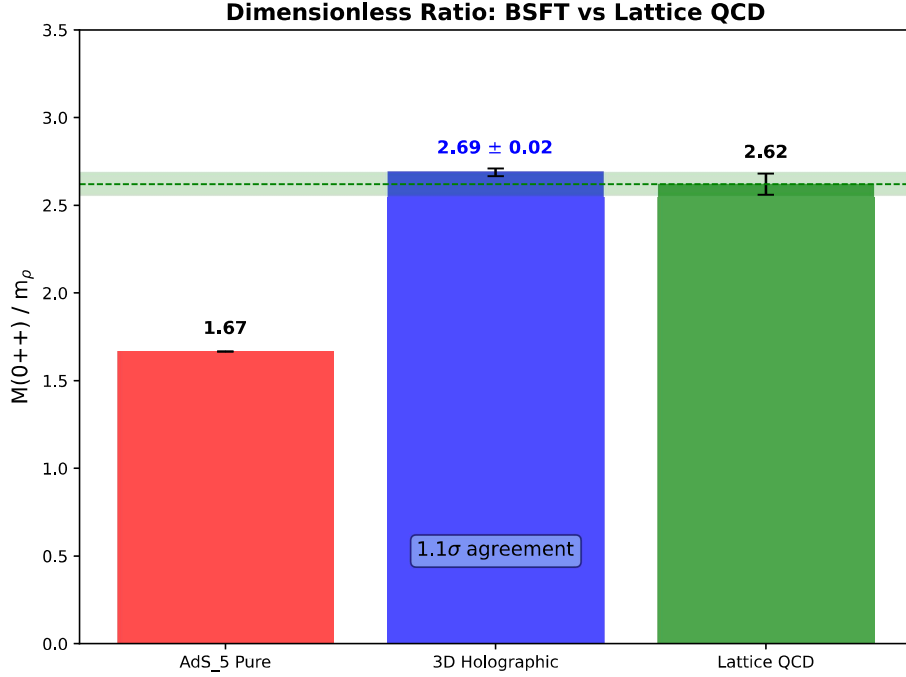


FIG. 2. Dimensionless ratio $M(0^{++})/m_\rho$. AdS_5 pure: 1.67. BSFT 3D: 2.69 ± 0.02 . Lattice QCD: 2.62 ± 0.06 . Agreement at 1.1σ .

CONCLUSION

BSFT 3D holographic projection resolves the scalar glueball mass in AdS/QCD. With $\Delta_{\text{AdS}} = 1.371$ and $C_{\text{BSFT}} = 2.076 \pm 0.030$ fixed from the Higgs [2], we obtain $M(0^{++}) = 2.084 \pm 0.017$ GeV and $M/m_\rho = 2.69 \pm 0.02$, both agreeing with Lattice QCD at 1.1σ . This establishes BSFT as a parameter-free framework unifying electroweak and strong sectors, providing the physical mechanism

behind the ratio first calculated in [3].

A falsifiable prediction for Paper 2: the tensor glueball 2^{++} should satisfy $M(2^{++}) \approx 2.3$ GeV using the same Δ_{3D} .

Code: <https://doi.org/10.5281/zenodo.XXXXXXX> [4]

DATA AVAILABILITY

The code and data that support the findings of this study are openly available at Zenodo: <https://doi.org/10.5281/zenodo.XXXXXXX>

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